TERM 2

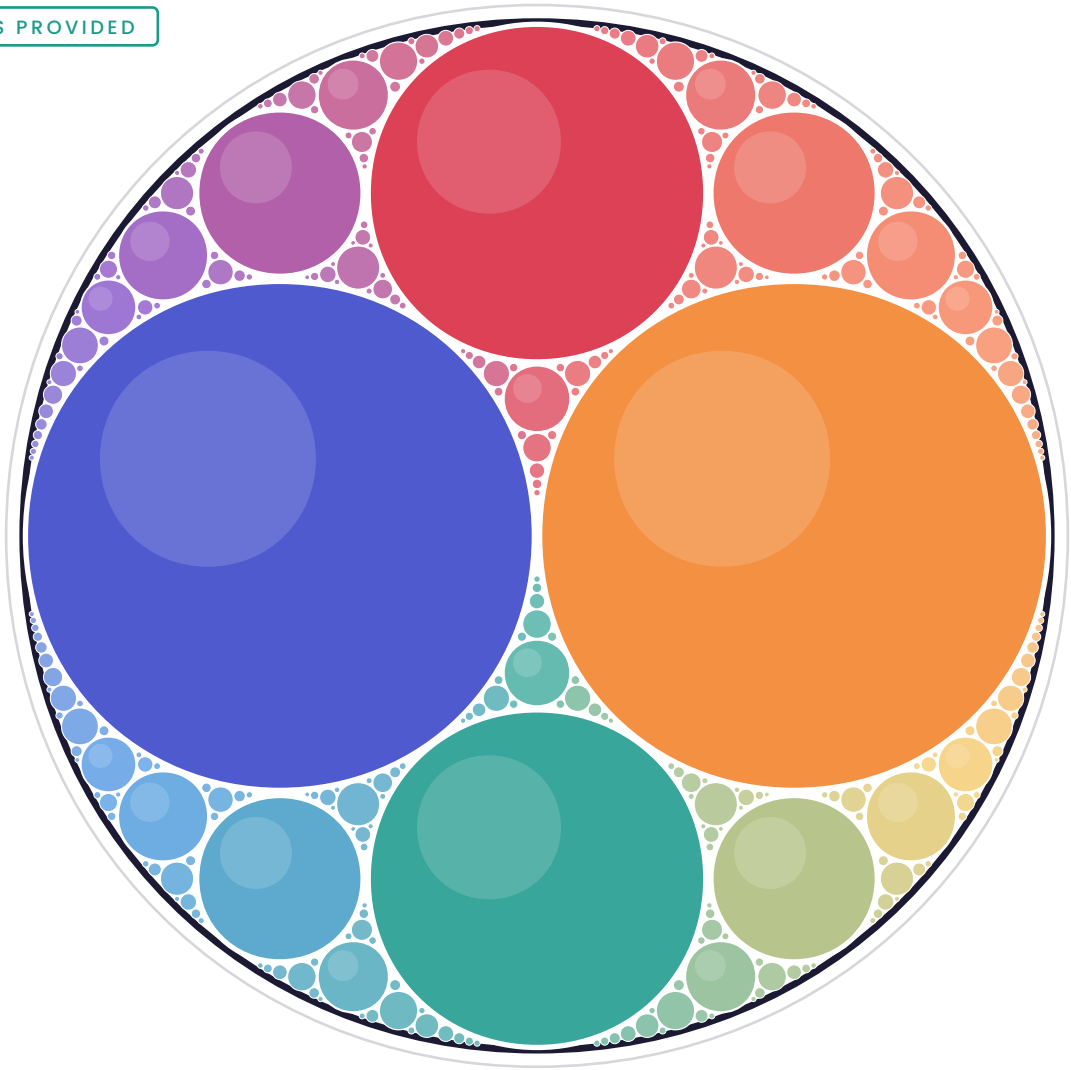
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
each circle fills a gap between three touching circles; every curvature (1 ÷ radius) is an integer

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

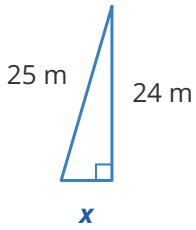
TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

2 A surf shop buys a cap for $20 and sells it for $21. Find the profit as a percentage of the cost price.

FORMULA Profit (or loss) % = profit (or loss) ÷ cost price × 100%

COPY

SUBSTITUTE

ANSWER

3 Find the mode of these scores.

Scores	9	4	5	1	9	9	6	3	5

FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

4 Which has the larger area: a 10 m by 3 m rectangle or a square with sides of 6 m? Explain. EXPLAIN

FORMULA $A = lw$ and $A = s^2$

COPY

WORKING

EXPLAIN

5 Calculate.

$\frac{1}{3}$ of 6 L

FORMULA Fraction of a quantity = fraction × quantity

COPY

SUBSTITUTE

ANSWER

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MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Increase $150 by 10%.

METHOD Find the percentage of the amount, then add it on

RULE

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$9^2 \times 9^5$$

FORMULA $a^m \times a^n = a^{m+n}$

COPY

SUBSTITUTE

ANSWER

3 Find the area of this rectangle.



FORMULA $A = lw$

COPY

SUBSTITUTE

ANSWER

4 The probability that the bus is late is $\frac{1}{7}$. Find the probability that the bus is not late.

FORMULA P(not E) = 1 − P(E)

COPY

SUBSTITUTE

ANSWER

5 Solve.

$$8 + 7a = 29$$

METHOD Do the same to both sides (use the inverse operation)

RULE

WORKING

ANSWER

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MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

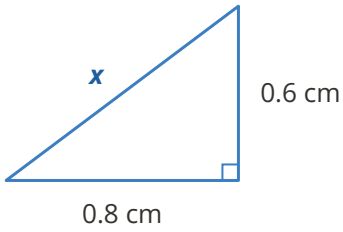
TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

2 50% of the students at a school walk to school. That is 60 students. How many students are at the school?

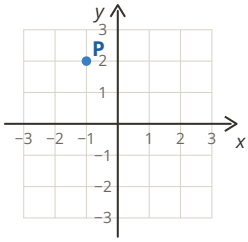
METHOD Unitary method: find 1% (or 10%) first, then multiply to get 100%

RULE

WORKING

ANSWER

3 Write the coordinates of the point P.



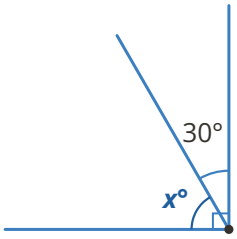
METHOD Coordinates (x, y): across first, then up or down

RULE

WORKING

ANSWER

4 Find the value of x.



FORMULA Complementary angles add to 90°

COPY

SUBSTITUTE

ANSWER

5 Round to 2 decimal places.

3.4672

METHOD Look at the next digit: 5 or more, round up; 4 or less, leave it

RULE

WORKING

ANSWER

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MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$7^4 \div 7^2$$

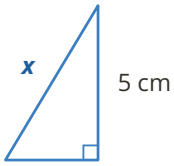
FORMULA $a^m \div a^n = a^{m-n}$

COPY

SUBSTITUTE

ANSWER

2 Find the hypotenuse, x, correct to 1 decimal place.



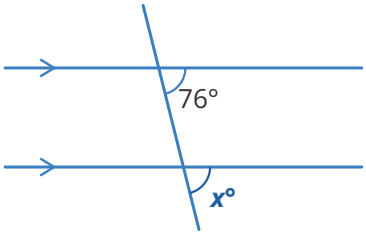
FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



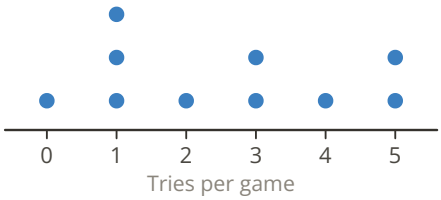
FORMULA Corresponding angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

4 The dot plot shows the tries a rugby league team scored in each game. Find the range.



FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

5 Calculate.

30% of 140 m

FORMULA Percentage of a quantity = percentage ÷ 100 × quantity

COPY

SUBSTITUTE

ANSWER

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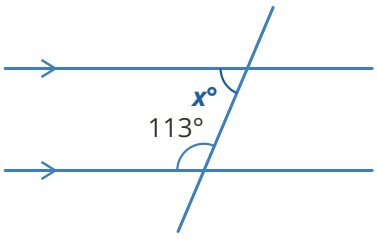
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5
1 Between which two whole numbers does this square root lie? $\sqrt{111}$	METHOD Find the square numbers just below and just above the number RULE WORKING ANSWER	
2 The price of a beach umbrella is $70 before GST. How much is the GST (10%)?	FORMULA GST = 10% of the price; price incl. GST = price + GST COPY SUBSTITUTE ANSWER	
3 Find the value of *x*. 	FORMULA Co-interior angles on parallel lines add to 180° COPY SUBSTITUTE ANSWER	
4 A bag holds 2 yellow and 6 green marbles. One is chosen at random. Is it more likely than not to be yellow? Explain. EXPLAIN	FORMULA P(E) = number of favourable outcomes ÷ total number of outcomes COPY WORKING EXPLAIN	
5 Calculate. 5.19×10	METHOD × 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right RULE WORKING ANSWER	

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MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

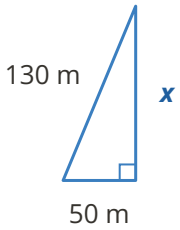
TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

2 The number of nippers changed from 120 to 180. Find the percentage increase.

FORMULA Percentage change = change ÷ original amount × 100%

COPY

SUBSTITUTE

ANSWER

3 Solve.

$$k - 5 = 12$$

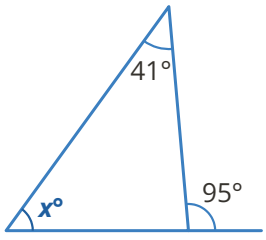
METHOD Do the same to both sides (use the inverse operation)

RULE

WORKING

ANSWER

4 Find the value of x.



FORMULA Exterior angle = sum of the 2 interior opposite angles

COPY

SUBSTITUTE

ANSWER

5 A netball shooter scored 30 goals from 50 shots. What percentage of the shots were goals?

FORMULA Percentage = part ÷ whole × 100%

COPY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 A triangle has sides 6 m, 8 m and 10 m. Is it right-angled?

FORMULA Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

COPY

SUBSTITUTE

ANSWER

2 Simplify. Write your answer as a single power.

$$(4^4)^3$$

FORMULA $(a^m)^n = a^{mn}$

COPY

SUBSTITUTE

ANSWER

3 Write as a recurring decimal (use dot notation).

$$\frac{2}{3}$$

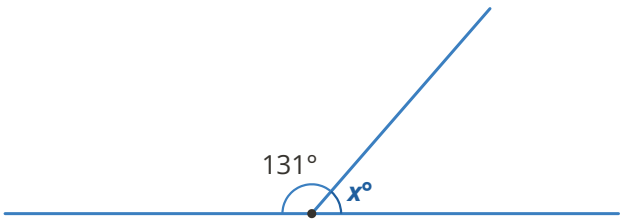
METHOD Numerator ÷ denominator; put a dot over each repeating digit

RULE

WORKING

ANSWER

4 Find the value of x.



FORMULA Angles on a straight line add to 180°

COPY

SUBSTITUTE

ANSWER

5 The stem-and-leaf plot shows the ages of people at a beach. Find the median.

2	0 1 3
3	0 3 4
4	0 6 6

Key: 2 | 0 = 20

FORMULA Median = middle score when the scores are in order

COPY

SUBSTITUTE

ANSWER

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MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

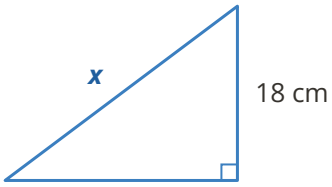
TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

2 A surfboard costs $660. It is on sale at 40% off. Find the sale price.

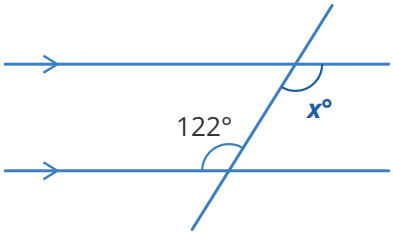
METHOD Discount = % × price; sale price = price − discount

RULE

WORKING

ANSWER

3 Find the value of x.



FORMULA Alternate angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

4 A spinner was spun 100 times. Find the observed probability of green.

Colour	Red	Blue	Green
Frequency	23	35	42

FORMULA Observed probability = frequency ÷ number of trials

COPY

SUBSTITUTE

ANSWER

5 The point $(3, k)$ lies on the line $y = x - 4$. Find the value of k.

METHOD A point lies on a line if its coordinates make the rule true

RULE

WORKING

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

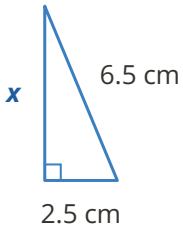
TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

2 Decrease 120 km by 40%.

METHOD Find the percentage of the amount, then take it away

RULE

WORKING

ANSWER

3 Simplify.

$$2 \times 3x$$

METHOD Multiply the numbers, then the pronumerals (in alphabetical order)

RULE

WORKING

ANSWER

4 Find the range of these waves caught by nippers.

Waves	4	1	5	3	1

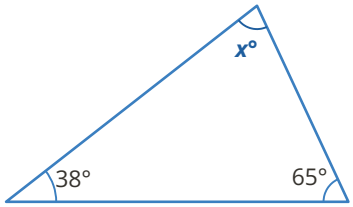
FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

5 Find the value of x. Give a reason. REASON



FORMULA Angle sum of a triangle = 180°

COPY

SUBSTITUTE

REASON

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Grace expanded $2(m + 2)$ and got $2m + 2$. Explain the error. EXPLAIN

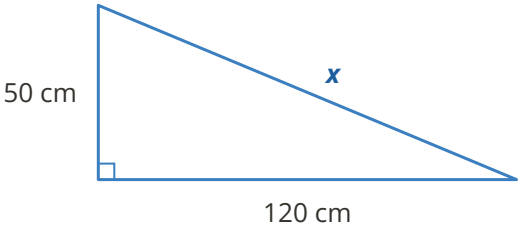
FORMULA $a(b + c) = ab + ac$

COPY

WORKING

EXPLAIN

2 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

3 Convert 3 cm^2 to mm^2.

FORMULA 1 cm^2 = 100 mm^2, 1 m^2 = 10 000 cm^2, 1 ha = 10 000 m^2

COPY

SUBSTITUTE

ANSWER

4 Evaluate.

$$10^2 + 2^4$$

FORMULA $a^n = a \times a \times \ldots \times a$ (n factors)

COPY

SUBSTITUTE

ANSWER

5 Find the mean of these goal tallies.

Goals	3	19	9	9

FORMULA Mean = sum of scores ÷ number of scores

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Factorise.

$$18y + 27$$

METHOD Take out the highest common factor (HCF); check by expanding

RULE

WORKING

ANSWER

2 Evaluate.

$$2^2 + 2^0$$

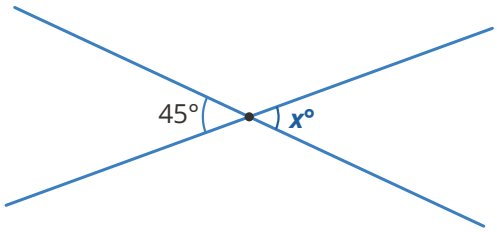
FORMULA $a^0 = 1$ (any non-zero number to the power 0 is 1)

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



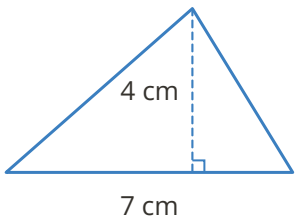
FORMULA Vertically opposite angles are equal

COPY

SUBSTITUTE

ANSWER

4 Find the area of this triangle.



FORMULA $A = \frac{1}{2}bh$

COPY

SUBSTITUTE

ANSWER

5 Write as a percentage.

$$\frac{3}{4}$$

METHOD Fraction to percentage: fraction × 100%

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand and simplify.

$$5(m + 4) + 6$$

METHOD Expand the brackets, then collect like terms

RULE

WORKING

ANSWER

2 A triangle has sides 10 m, 24 m and 26 m. Is it right-angled?

FORMULA Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

COPY

SUBSTITUTE

ANSWER

3 Find the median of these try tallies.

Tries	6	0	1	2	3

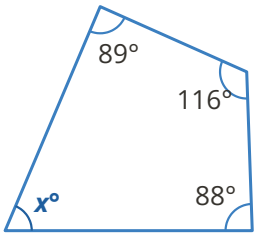
FORMULA Median = middle score when the scores are in order (2 middles: halfway)

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Angle sum of a quadrilateral = 360°

COPY

SUBSTITUTE

ANSWER

5 Evaluate.

$$\sqrt{25}$$

METHOD $\sqrt{a}$ is the number that squares to a; $\sqrt[3]{a}$ is the number that cubes to a

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify.

$$24x^2 \div 4x$$

METHOD Divide the numbers, then cancel common pronumerals ($a^2 \div a = a$)

RULE

WORKING

ANSWER

2 A surf shop buys a T-shirt for $20 and sells it for $25. Find the profit as a percentage of the cost price.

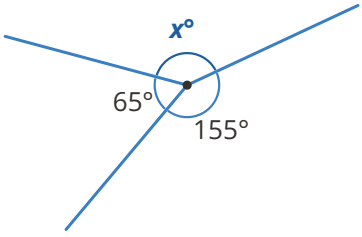
FORMULA Profit (or loss) % = profit (or loss) ÷ cost price × 100%

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



FORMULA Angles at a point add to 360°

COPY

SUBSTITUTE

ANSWER

4 A spinner has 10 equal sections. 6 of them are green. Find the probability of landing on green.

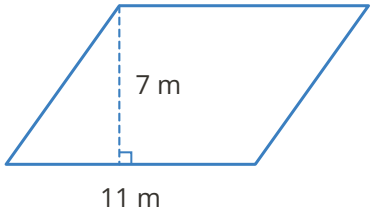
FORMULA P(E) = number of favourable outcomes ÷ total number of outcomes

COPY

SUBSTITUTE

ANSWER

5 Find the area of this parallelogram.



FORMULA $A = bh$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$4^3 \times 4^2$$

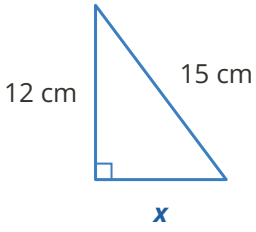
FORMULA $a^m \times a^n = a^{m+n}$

COPY

SUBSTITUTE

ANSWER

2 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

3 Is $m = 11$ the solution of this equation? Show how you know. EXPLAIN

$$6m + 14 = 80$$

METHOD Check a solution by substituting it into both sides

RULE

WORKING

EXPLAIN

4 Evaluate.

$$5 \times 8 - 3 \times 10$$

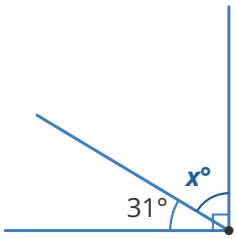
METHOD Brackets, then × and ÷ (left to right), then + and − (left to right)

RULE

WORKING

ANSWER

5 Find the value of x.



FORMULA Complementary angles add to 90°

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify.

$$5m^2 + 5m - 3m^2 + 4m$$

METHOD Like terms have exactly the same pronumerals and powers

RULE

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$5^8 \div 5^4$$

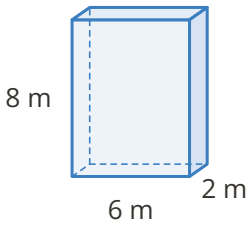
FORMULA $a^m \div a^n = a^{m-n}$

COPY

SUBSTITUTE

ANSWER

3 Find the volume of this rectangular prism.



FORMULA $V = lwh$

COPY

SUBSTITUTE

ANSWER

4 Write 52 as a product of its prime factors.

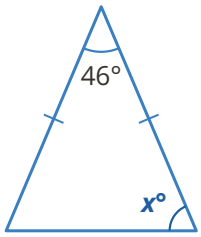
METHOD Use a factor tree: keep splitting until every branch ends in a prime

RULE

WORKING

ANSWER

5 Find the value of x.



FORMULA Isosceles triangle: base angles are equal; angle sum = 180°

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Write 80 as a product of prime factors in index form.

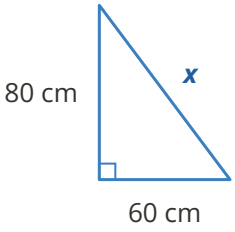
METHOD Factor tree to primes, then write repeated primes as powers

RULE

WORKING

ANSWER

2 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

3 Complete the table for $y = x + 1$.

x	−1	0	1	2
y	0		2	

METHOD Substitute each x-value into the rule

RULE

WORKING

ANSWER

4 Evaluate.

$$2 \times (-9)$$

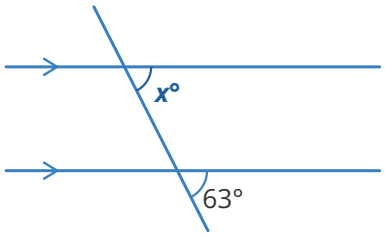
METHOD Same signs: answer positive. Different signs: answer negative

RULE

WORKING

ANSWER

5 Find the value of x.



FORMULA Corresponding angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Increase 80 kg by 20%.

METHOD Find the percentage of the amount, then add it on

RULE

WORKING

ANSWER

2 Find the value of this expression when $p = -2$ and $q = 2$.

$$pq + 5$$

METHOD Replace each pronumeral with its value (use brackets)

RULE

WORKING

ANSWER

3 Find the mode of these scores.

Scores	8	8	3	9	8	8	3

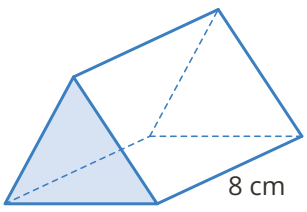
FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

4 The shaded cross-section has an area of 25 cm^2. Find the volume of this prism.



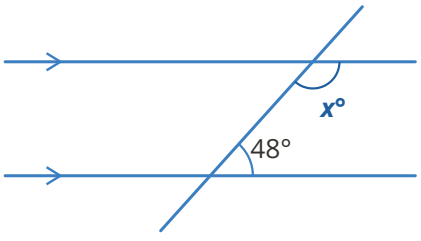
FORMULA $V = Ah$

COPY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Co-interior angles on parallel lines add to 180°

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand.

$$3(4m - 9)$$

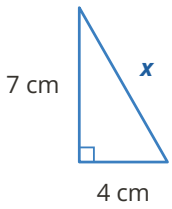
FORMULA $a(b + c) = ab + ac$

COPY

SUBSTITUTE

ANSWER

2 Find the hypotenuse, x, correct to 1 decimal place.



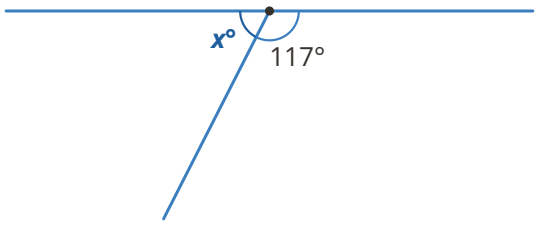
FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



FORMULA Angles on a straight line add to 180°

COPY

SUBSTITUTE

ANSWER

4 The probability that the flight is delayed is 15%. Find the probability that the flight is not delayed.

FORMULA P(not E) = 1 − P(E)

COPY

SUBSTITUTE

ANSWER

5 Mia says $11^2 = 22$. Explain her mistake and give the correct value. EXPLAIN

FORMULA $a^n = a \times a \times \ldots \times a$ (n factors)

RULE

WORKING

EXPLAIN

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Is 5% of \$80 the same as 80% of \$5? Explain. EXPLAIN

METHOD Percentage of a quantity = percentage ÷ 100 × quantity

RULE

WORKING

EXPLAIN

2 A triangle has sides 6 cm, 7 cm and 9 cm. Is it right-angled?

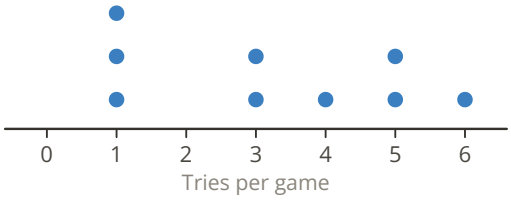
FORMULA Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

COPY

SUBSTITUTE

ANSWER

3 The dot plot shows the tries a rugby league team scored in each game. Find the mode.



FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

4 $\frac{2}{5}$ of the 20 students in a class walk to school. How many students is that?

FORMULA Fraction of a quantity = fraction × quantity

COPY

SUBSTITUTE

ANSWER

5 Simplify.

$$8p - 4p + 8$$

METHOD Only like terms (the same pronumerals) can be added or subtracted

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Factorise.

$$32m + 24$$

METHOD Take out the highest common factor (HCF); check by expanding

RULE

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$(4^4)^4$$

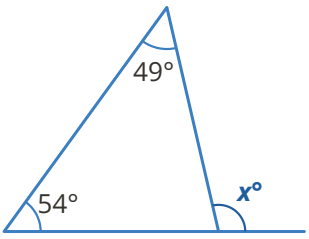
FORMULA $(a^m)^n = a^{mn}$

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



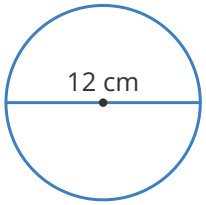
FORMULA Exterior angle = sum of the 2 interior opposite angles

COPY

SUBSTITUTE

ANSWER

4 Find the circumference of this circle, correct to 1 decimal place.



FORMULA $C = \pi d$

COPY

SUBSTITUTE

ANSWER

5 Calculate.

80% of $240

FORMULA Percentage of a quantity = percentage ÷ 100 × quantity

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand and simplify.

$$3(n + 6) + 4$$

METHOD Expand the brackets, then collect like terms

RULE

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$9^4 \times 9^5$$

FORMULA $a^m \times a^n = a^{m+n}$

COPY

SUBSTITUTE

ANSWER

3 Find the range of these netball scores.

Points	7	7	10	19	15

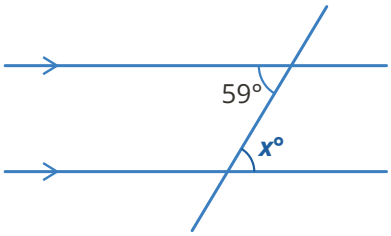
FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



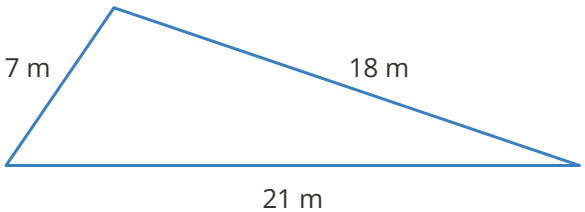
FORMULA Alternate angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

5 Find the perimeter of this triangle.



METHOD Perimeter = the sum of all the side lengths

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand.

$$-4(3y + 6)$$

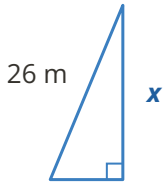
FORMULA $a(b + c) = ab + ac$

COPY

SUBSTITUTE

ANSWER

2 Find the length of the shorter side, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

3 A die was rolled 60 times. Find the observed probability of rolling a 6.

Number	1	2	3	4	5	6
Frequency	13	8	12	9	7	11

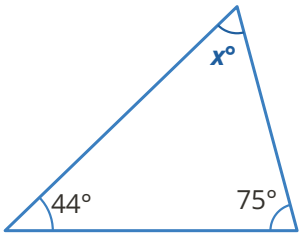
FORMULA Observed probability = frequency ÷ number of trials

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Angle sum of a triangle = 180°

COPY

SUBSTITUTE

ANSWER

5 Calculate.

$$9.6 \div 3$$

METHOD Divide as normal; keep the decimal point in line

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 25% of the students at a school walk to school. That is 40 students. How many students are at the school?

METHOD Unitary method: find 1% (or 10%) first, then multiply to get 100%

RULE

WORKING

ANSWER

2 Factorise.

$$12x + 3$$

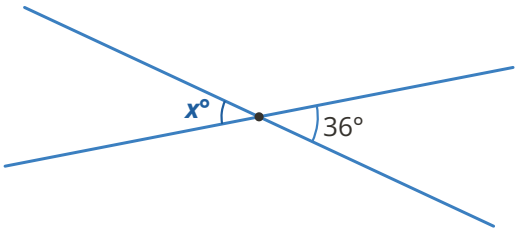
METHOD Take out the highest common factor (HCF); check by expanding

RULE

WORKING

ANSWER

3 Find the value of x.



FORMULA Vertically opposite angles are equal

COPY

SUBSTITUTE

ANSWER

4 Find the mean of these try tallies.

Tries	6	0	3	7	4

FORMULA Mean = sum of scores ÷ number of scores

COPY

SUBSTITUTE

ANSWER

5 Which has the larger area: a 10 m by 4 m rectangle or a square with sides of 6 m? Explain. EXPLAIN

FORMULA $A = lw$ and $A = s^2$

COPY

WORKING

EXPLAIN

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

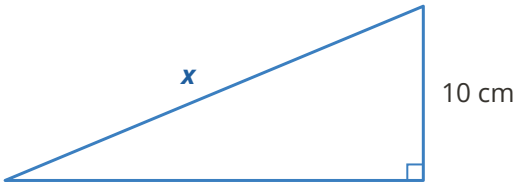
TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the hypotenuse, x.



FORMULA $c^2 = a^2 + b^2$ (c is the hypotenuse)

COPY

SUBSTITUTE

ANSWER

2 Simplify.

$$2x \times 4x$$

METHOD Multiply the numbers, then the pronumerals ($a \times a = a^2$)

RULE

WORKING

ANSWER

3 A letter is chosen at random from this word. Find P(H).

D O L P H I N

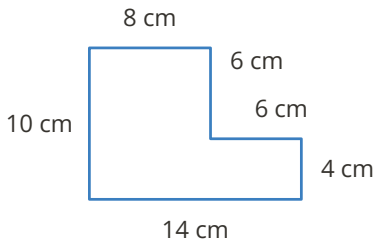
FORMULA P(E) = number of favourable outcomes ÷ total number of outcomes

COPY

SUBSTITUTE

ANSWER

4 Find the area of this shape.



METHOD Split the shape into rectangles (and triangles), then add their areas

RULE

WORKING

ANSWER

5 Calculate. Give your answer in simplest form.

$$\frac{1}{7} + \frac{1}{7}$$

METHOD Same denominators: add or subtract the numerators, then simplify

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The price of a bike helmet is $110 before GST. Add 10% GST to find the price including GST.

FORMULA GST = 10% of the price; price incl. GST = price + GST

COPY

SUBSTITUTE

ANSWER

2 Solve.

$$6x + 3 = 3x + 12$$

METHOD Move the pronumerals to one side first, then solve

RULE

WORKING

ANSWER

3 The Kingscliff touch footy team won 34 of its 40 games. What percentage of its games did it win?

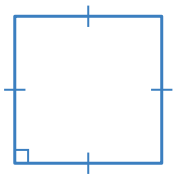
FORMULA Percentage = part ÷ whole × 100%

COPY

SUBSTITUTE

ANSWER

4 Find the area of this square.



15 m

FORMULA $A = s^2$

COPY

SUBSTITUTE

ANSWER

5 The stem-and-leaf plot shows the ages of people at a beach. Find the range.

1	0 9
2	2 5 6
3	3 3 3 7

Key: 1 | 0 = 10

FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Use the formula to find l when $A = 32$ and $w = 8$.

FORMULA $A = lw$

COPY

SUBSTITUTE

ANSWER

2 Between which two whole numbers does this square root lie?

$$\sqrt{93}$$

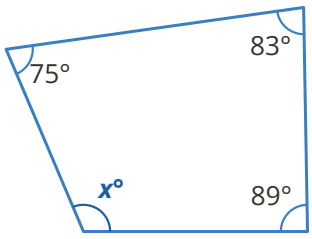
METHOD Find the square numbers just below and just above the number

RULE

WORKING

ANSWER

3 Find the value of x.



FORMULA Angle sum of a quadrilateral = 360°

COPY

SUBSTITUTE

ANSWER

4 Calculate.

$$260 \div 1000$$

METHOD × 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right

RULE

WORKING

ANSWER

5 Find the median of these heights.

cm	133	145	129	131	151

FORMULA Median = middle score when the scores are in order (2 middles: halfway)

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$a^2 + 9 = 45$$

METHOD $x^2 = c$ has 2 solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

RULE

WORKING

ANSWER

2 Expand.

$$-2(x - 5)$$

FORMULA $a(b + c) = ab + ac$

COPY

SUBSTITUTE

ANSWER

3 Write as a fraction in simplest form.

$$15\%$$

METHOD Percentage to fraction: write over 100, then simplify

RULE

WORKING

ANSWER

4 For these scores, which is larger: the mean or the median? Explain. EXPLAIN

Scores	4	7	6	23	5

FORMULA Mean = sum of scores ÷ number of scores

COPY

WORKING

EXPLAIN

5 Convert 250 cm^3 to millilitres.

FORMULA 1 cm^3 = 1 mL, 1 L = 1000 mL, 1 m^3 = 1000 L

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Leo thinks of a number, multiplies it by 5 and subtracts 4. The answer is 36. Write and solve an equation to find the number.

METHOD Let n be the number; write an equation, then solve it

RULE

WORKING

ANSWER

2 Factorise.

$$4m + 14$$

METHOD Take out the highest common factor (HCF); check by expanding

RULE

WORKING

ANSWER

3 Round to 1 decimal place.

0.7349

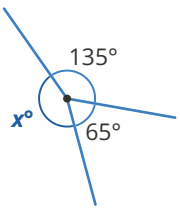
METHOD Look at the next digit: 5 or more, round up; 4 or less, leave it

RULE

WORKING

ANSWER

4 Find the value of x. Give a reason. REASON



FORMULA Angles at a point add to 360°

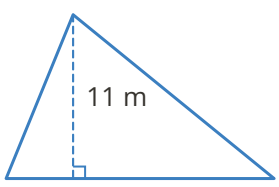
COPY

SUBSTITUTE

REASON

ANSWER

5 Find the area of this triangle.



FORMULA $A = \frac{1}{2}bh$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve. Give your answer as a fraction or a decimal.

$$2x + 6 = 11$$

METHOD Do the same to both sides (use the inverse operation)

RULE

WORKING

ANSWER

2 A triangle has sides 2 m, 3 m and 4 m. Is it right-angled?

FORMULA Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

COPY

SUBSTITUTE

ANSWER

3 Evaluate.

$$\sqrt[3]{1000}$$

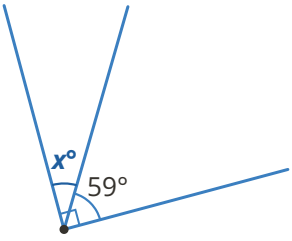
METHOD $\sqrt{a}$ is the number that squares to a; $\sqrt[3]{a}$ is the number that cubes to a

RULE

WORKING

ANSWER

4 Find the value of x.



FORMULA Complementary angles add to 90°

COPY

SUBSTITUTE

ANSWER

5 Find the mode of these scores.

Scores	1	3	8	6	1	7	6	1	8

FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 2 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$7x + 2 = 5x + 8$$

METHOD Move the pronumerals to one side first, then solve

RULE

WORKING

ANSWER

2 Expand and simplify.

$$4(n + 5) + 1$$

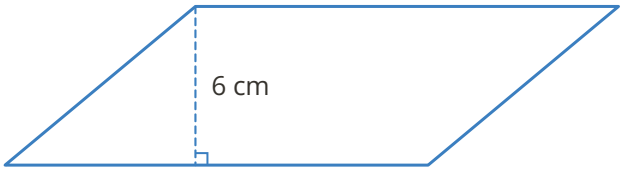
METHOD Expand the brackets, then collect like terms

RULE

WORKING

ANSWER

3 Find the area of this parallelogram.



16 cm

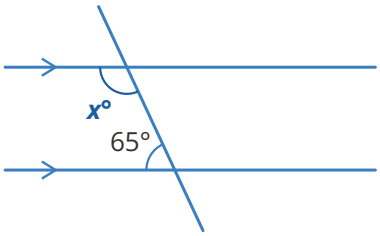
FORMULA $A = bh$

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Co-interior angles on parallel lines add to 180°

COPY

SUBSTITUTE

ANSWER

5 Evaluate.

$$(12 - 8) \times 11$$

METHOD Brackets, then × and ÷ (left to right), then + and − (left to right)

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Formula · Reference 1

TERM 2 Every formula and rule used in this booklet.

NUMBER

FRACTIONS AND PERCENTAGES OF A QUANTITY

Fraction of a quantity = fraction × quantity

Percentage of a quantity = percentage ÷ 100 × quantity

Percentage = part ÷ whole × 100%

CONVERTING FRACTIONS, DECIMALS AND PERCENTAGES

Fraction to percentage: fraction × 100%

Percentage to fraction: write over 100, then simplify

POWERS AND ROOTS

$a^n = a \times a \times \ldots \times a$ (n factors)

$\sqrt{a}$ is the number that squares to a; $\sqrt[3]{a}$ is the number that cubes to a

ADDING, SUBTRACTING AND MULTIPLYING FRACTIONS

Same denominators: add or subtract the numerators, then simplify

INTEGERS

Same signs: answer positive. Different signs: answer negative

MULTIPLYING AND DIVIDING BY 10, 100, 1000

× 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right

OPERATIONS WITH DECIMALS

Divide as normal; keep the decimal point in line

ORDER OF OPERATIONS

Brackets, then × and ÷ (left to right), then + and − (left to right)

PRIME FACTORS (FACTOR TREE)

Use a factor tree: keep splitting until every branch ends in a prime

RECURRING DECIMALS

Numerator ÷ denominator; put a dot over each repeating digit

ROUNDING DECIMALS

Look at the next digit: 5 or more, round up; 4 or less, leave it

INDEX LAWS · YEAR 8

$a^m \times a^n = a^{m+n}$

$a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$

$a^0 = 1$ (any non-zero number to the power 0 is 1)

PERCENTAGE INCREASE AND DECREASE · YEAR 8

Find the percentage of the amount, then add it on

Percentage change = change ÷ original amount × 100%

Discount = % × price; sale price = price − discount

Find the percentage of the amount, then take it away

PROFIT, LOSS AND GST · YEAR 8

Profit (or loss) % = profit (or loss) ÷ cost price × 100%

GST = 10% of the price; price incl. GST = price + GST

PRIME FACTORS IN INDEX FORM · YEAR 8

Factor tree to primes, then write repeated primes as powers

THE UNITARY METHOD · YEAR 8

Unitary method: find 1% (or 10%) first, then multiply to get 100%

ALGEBRA AND EQUATIONS

LINEAR RELATIONSHIPS

Coordinates (x, y): across first, then up or down

A point lies on a line if its coordinates make the rule true

Substitute each x-value into the rule

MULTIPLYING AND DIVIDING TERMS

Multiply the numbers, then the pronumerals (in alphabetical order)

Divide the numbers, then cancel common pronumerals ($a^2 \div a = a$)

Multiply the numbers, then the pronumerals ($a \times a = a^2$)

LIKE TERMS

Like terms have exactly the same pronumerals and powers

Only like terms (the same pronumerals) can be added or subtracted

SOLVING EQUATIONS

Do the same to both sides (use the inverse operation)

Check a solution by substituting it into both sides

EQUATION PROBLEMS · YEAR 8

Let n be the number; write an equation, then solve it

EQUATIONS OF THE FORM $AX^2 = C$ · YEAR 8

$x^2 = c$ has 2 solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

EXPANDING AND SIMPLIFYING · YEAR 8

Expand the brackets, then collect like terms

EXPANDING BRACKETS · YEAR 8

$a(b + c) = ab + ac$

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Formula · Reference 2

TERM 2 Every formula and rule used in this booklet (continued).

ALGEBRA AND EQUATIONS (CONTINUED)

FACTORISING · YEAR 8
Take out the highest common factor (HCF); check by expanding

PRONUMERALS ON BOTH SIDES · YEAR 8
Move the pronumerals to one side first, then solve

SUBSTITUTION · YEAR 8
Replace each pronumeral with its value (use brackets)

MEASUREMENT

AREA OF RECTANGLES AND SQUARES
$A = lw$ and $A = s^2$

$A = lw$

$A = s^2$

UNITS OF AREA, VOLUME AND CAPACITY
1 cm^2 = 100 mm^2, 1 m^2 = 10 000 cm^2, 1 ha = 10 000 m^2

1 cm^3 = 1 mL, 1 L = 1000 mL, 1 m^3 = 1000 L

VOLUME OF PRISMS
$V = lwh$

$V = Ah$

AREA OF A COMPOSITE SHAPE
Split the shape into rectangles (and triangles), then add their areas

AREA OF A PARALLELOGRAM
$A = bh$

AREA OF A TRIANGLE
$A = \frac{1}{2}bh$

CIRCUMFERENCE OF A CIRCLE
$C = \pi d$

PERIMETER
Perimeter = the sum of all the side lengths

PYTHAGORAS' THEOREM · YEAR 8
$c^2 = a^2 + b^2$ (c is the hypotenuse)

Find the square numbers just below and just above the number

Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

GEOMETRY: ANGLES

ANGLES AT A POINT AND ON A LINE
Complementary angles add to 90°

Angles on a straight line add to 180°

Vertically opposite angles are equal

Angles at a point add to 360°

TRIANGLES AND QUADRILATERALS
Exterior angle = sum of the 2 interior opposite angles

Angle sum of a triangle = 180°

Angle sum of a quadrilateral = 360°

Isosceles triangle: base angles are equal; angle sum = 180°

ANGLES ON PARALLEL LINES
Corresponding angles on parallel lines are equal

Co-interior angles on parallel lines add to 180°

Alternate angles on parallel lines are equal

STATISTICS AND PROBABILITY

MEAN, MEDIAN, MODE AND RANGE
Mode = the most common score

Range = highest score − lowest score

Median = middle score when the scores are in order

Mean = sum of scores ÷ number of scores

Median = middle score when the scores are in order (2 middles: halfway)

PROBABILITY
P(not E) = 1 − P(E)

P(E) = number of favourable outcomes ÷ total number of outcomes

Observed probability = frequency ÷ number of trials

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

My · Tracker

TERM 2 Write your score after each session.

NAME _____

DATE _____

WEEK	SESSION 1	SESSION 2	SESSION 3	WEEK TOTAL
Week 1	/ 5	/ 5	/ 5	/ 15
Week 2	/ 5	/ 5	/ 5	/ 15
Week 3	/ 5	/ 5	/ 5	/ 15
Week 4	/ 5	/ 5	/ 5	/ 15
Week 5	/ 5	/ 5	/ 5	/ 15
Week 6	/ 5	/ 5	/ 5	/ 15
Week 7	/ 5	/ 5	/ 5	/ 15
Week 8	/ 5	/ 5	/ 5	/ 15
Week 9	/ 5	/ 5	/ 5	/ 15
Week 10	/ 5	/ 5	/ 5	/ 15
TERM TOTAL				**/ 150**

OUTCOMES IN THIS BOOKLET · NSW MATHEMATICS K–10 (2022), STAGE 4

MA4-INT-C-01	compares, orders and calculates with integers to solve problems	NCM 7
MA4-FRC-C-01	represents and operates with fractions, decimals and percentages to solve problems	NCM 7, NCM 8
MA4-IND-C-01	operates with primes and roots, positive-integer and zero indices involving numerical bases and establishes the relevant index laws	NCM 7, NCM 8
MA4-ALG-C-01	generalises number properties to operate with algebraic expressions including expansion and factorisation	NCM 7, NCM 8
MA4-EQU-C-01	solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$	NCM 7, NCM 8
MA4-LIN-C-01	creates and displays number patterns and finds graphical solutions to problems involving linear relationships	NCM 7
MA4-LEN-C-01	applies knowledge of the perimeter of plane shapes and the circumference of circles to solve problems	NCM 7
MA4-ARE-C-01	applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems	NCM 7
MA4-VOL-C-01	applies knowledge of volume and capacity to solve problems involving right prisms and cylinders	NCM 7
MA4-PYT-C-01	applies Pythagoras' theorem to solve problems in various contexts	NCM 8
MA4-ANG-C-01	applies angle relationships to solve problems, including those related to transversals on sets of parallel lines	NCM 7
MA4-GEO-C-01	identifies and applies the properties of triangles and quadrilaterals to solve problems	NCM 7
MA4-DAT-C-02	analyses simple datasets using measures of centre, range and shape of the data	NCM 7
MA4-PRO-C-01	solves problems involving the probabilities of simple chance experiments	NCM 7
MAO-WM-01	Working mathematically: the weekly reasoning questions (marked EXPLAIN or REASON) ask students to explain, check or justify.	

Year 8 units in these warm-ups (each starts once its teaching ends in the 2027 Year 8 mixed ability program): Rational numbers, percentages & financial maths from T1 Wk 5; Index laws from T1 Wk 7; Pythagoras' theorem from T1 Wk 10; Algebra 2 from T2 Wk 4; Equations 2 from T2 Wk 9. All other questions revise Year 7 (NCM 7 Ch 1–11). Ratios, rates and time (NCM 7 Ch 12) are held back until Year 8 teaches them.